

FINAL PROJECT REPORT

Smartphone Usage, Digital Addiction & Well-Being Analytics

An interactive Microsoft Power BI study of screen time, addiction levels, sleep and stress

Submitted by

C. Yamini

Master of Computer Applications (MCA)

Sri Padmavati Mahila Visvavidyalayam, Tirupati

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1. Introduction

1.1. Project Overview

Smartphones are now central to studying, working and socialising, but constant notifications, endless feeds and app-opening habits have made screen addiction a growing behavioural concern. **Smartphone Usage, Digital Addiction & Well-Being Analytics** studies a dataset of **7,500 smartphone users (aged 18–35)** to understand how screen time, social media, gaming, notifications and app-opening habits relate to digital addiction, sleep, stress and study/work outcomes.

The raw CSV data was profiled with SQL, cleaned and enriched in Power BI (Power Query), and presented through a **five-page interactive Power BI report** made up of a KPI page, a visualization page, a dashboard and two analytical reports.

7,500 Users analysed	16 Source attributes	5 Report pages	51 Visuals built	8 Slicers
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1.2. Objectives

- **Measure usage:** daily screen time (7.50 h), weekend screen time (9.24 h), social media, gaming, notifications and app opens.
- **Classify addiction:** segment users into None, Mild, Moderate and Severe levels; compare addicted and non-addicted users.
- **Assess well-being impact:** study how screen habits relate to sleep hours, stress level and academic/work impact.
- **Compare cohorts:** analyse behaviour by gender and age group (Under 20, 20–29, 30–39).
- **Find the drivers:** use Key Influencers and Decomposition Tree visuals to see what most raises addiction likelihood.
- **Support action:** give students, counsellors and advisors a clear dashboard for digital-detox planning.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Many people spend too much time on their smartphones. Constant notifications, chat alerts and endless scrolling pull them back to the phone again and again, which disturbs study, work and sleep. Users cannot see exactly where their time goes, so they cannot correct the habit. This project builds an analytics tool that makes those habits visible and identifies the factors behind addiction.

I am	A student or work-from-home professional who uses a smartphone heavily every day.
I'm trying to	Complete daily tasks on time, focus deeply on work or studies, and get a good night's sleep.
But	Constant phone alerts, endless scrolling feeds and addictive apps keep breaking my concentration.
Because	I cannot see exactly how much time I waste on my phone, and I have no tools to block these distractions.
Which makes me feel	Anxious, tired, upset about wasting time, and helpless against my phone habits.

Example scenarios: a student who cannot study for 3 hours without checking the phone (PS-1, Screen Fatigue) and a remote employee who cannot log off after 6 PM (PS-2, Remote Burnout).

2.2. Project Proposal (Proposed Solution)

Objective	Build an interactive Power BI dashboard to analyse smartphone screen time, measure digital addiction levels, and study their impact on sleep, stress and productivity.
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Scope	Collect usage data (screen time, social media, gaming, notifications, app opens), clean and enrich it, build the data model, and design multi-page visual reports.
Impact	Helps users and counsellors spot unhealthy habits early, understand effects on sleep and work, and build balanced digital habits.
Approach	Profile the data with SQL, clean and transform it in Power Query, add calculated columns, and present insights through interactive Power BI charts, slicers and AI visuals.
Key Features	Slicers for age group, gender, addiction level and stress level; four addiction levels (None, Mild, Moderate, Severe); Key Influencers and Decomposition Tree; charts linking screen hours to sleep and stress.
Resources	Intel Core i5 / Ryzen 5 (4+ cores), 8–16 GB RAM, 256–512 GB SSD; Power BI Desktop, Power Query, MySQL / SQL, Excel / CSV, Git.
Data	Smartphone_Usage.csv: 7,500 records with usage logs, sleep, stress and addiction indicators.

2.3. Initial Project Planning

The work was organised into three sprints with ten user stories and a total of **24 story points**. The table summarises the prioritised backlog that guided delivery of the final report.

Sprint	Epics	User Stories (Points, Priority)	Points
Sprint-1	Data Ingestion & Preprocessing; KPI Modelling	USN-01 Extract and clean usage metrics (3, High); USN-02 Relate demographics and well-being scores (2, High); USN-03 Create KPI calculations (3, High)	8
Sprint-2	Dashboard Overview & KPIs; Category Analytics; Addiction Classification	USN-04 Summary page with KPI cards and slicers (3, High); USN-05 Bar and donut charts by usage category (2, Medium); USN-06 Segment users by addiction level (3, High)	8
Sprint-3	Well-Being Analytics; Demographics; Testing & Theme; Documentation	USN-07 Screen time vs sleep and stress visuals (3, High); USN-08 Cross-filter by age and gender (2, Medium); USN-09 Theme, optimise and validate filters (2, Medium); USN-10 Final documentation (1, Low)	8

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The plan was to obtain a single, reliable usage log covering screen habits, notifications, sleep, stress and addiction indicators, and to keep it intact so that every finding can be traced back to the source file.

Source Name	Description	Format	Size	Access
smartphone_usage dataset	7,500 users with daily screen time, social media, gaming, work/study hours, sleep, notifications, app opens, weekend screen time, stress, academic/work impact and addiction level.	CSV	≈ 0.6 MB (7,500 rows × 16 columns)	Shared link (Google Drive)

Source location: https://drive.google.com/file/d/1jaS03ZC96lXiIMtm0BwRDaus_uU1sMX1/view

3.2. Data Quality Report

The dataset was checked with SQL and Power Query. It contains **no missing values and no duplicate records** (7,500 unique transaction and user IDs). The value “None” in the addiction_level column is a valid category (819 users), not a missing entry. The issues below concern usability for reporting and were resolved with derived columns.

Column / Area	Data Quality Issue	Severity	Resolution (derived column)
Age	Raw ages had no brackets for grouping.	Moderate	Age Group: Under 20 (799), 20–29 (4,132), 30–39 (2,569 users).
Addiction	Addiction stored only as a 0/1 code.	High	Addicted Status: 1 = “Addicted” (5,308), 0 = “Not Addicted” (2,192).
Screen time	No way to separate heavy from normal users.	High	High Screen Time User: above 6 h/day = “High Usage” (66.13%), else “Normal Usage”.
User count	No fixed total-user field for cards.	Moderate	User Count column carrying the 7,500 total.
Academic/work impact	Mixed impact values, hard to filter.	Moderate	Conditional column tagging High, Medium, Low impact.
Stress level	Stress needed simple categories.	Low	Grouped into Low, Medium and High Stress.
Sleep hours	Sleep not grouped, hiding low-sleep users.	Moderate	Low Sleep (under 6 h) / Normal Sleep (6 h+).

3.3. Data Exploration and Preprocessing

Stage	Work carried out in this project
Data Overview	7,500 rows × 16 columns, one row per user. Ranges: age 18–35; screen time 3–12 h; sleep 4.5–9 h; notifications 20–250; app opens 15–180.
Data Cleaning	SQL null checks returned no blanks and IDs had no duplicates; all values were in valid ranges, so no rows were removed or imputed.
Data Transformation	Column names were made readable (e.g. “daily screen time hours”) and the calculated and conditional columns from section 3.2 were added.
Data Type Conversion	Text: transaction_id, user_id, gender, stress_level, academic_work_impact, addiction_level. Whole number: age, notifications_per_day, app_opens_per_day, addicted_label. Decimal: all hour-based columns.
Column Splitting and Merging	No compound fields existed, so no splitting was needed; categories were derived from existing columns instead.
Data Modelling	The data is one flat table (smartphone_usage), so a single-table model was used with no relationships. Metrics use Power BI aggregations (Count, Average, Sum).
Save Processed Data	Steps were applied and loaded, and the model saved as smartphone.pbix for all five report pages.

4. Data Visualization

4.1. Framing Business Questions

Specific business questions were defined first so that each visual answers a clear need. Eight questions guided the visualization page.

#	Business Question	Visualization Used
1	How does daily screen time differ across genders?	Column chart: Average Daily Screen Time by Gender
2	How are users distributed across addiction severity levels?	Bar chart: Users by Addiction Level
3	What share of users is addicted vs not addicted?	Donut chart: Addicted vs Non-Addicted Users
4	Is daily screen time related to sleep duration?	Scatter chart: Daily Screen Time vs Sleep Hours
5	How does screen time vary across age cohorts?	Column chart: Average Daily Screen Time by Age Group
6	How does gaming compare with social media use?	Clustered column: Average Gaming vs Social Media Hours
7	How can users slice data by demographics and addiction level?	Slicers: Age Group, Gender, Addiction Level
8	How many users are captured in the study?	Card visual: Count of user id (7,500)

4.2. Developing Visualizations

KPI cards. Figure 1 shows the eleven KPI cards: 7,500 total users, 5,308 addicted users (count filtered to Addicted Status = Addicted), average age 26.57, and daily averages for screen time (7.50 h), social media (3.27 h), gaming (2.01 h), sleep (6.74 h), notifications (134.26), app opens (97.83), weekend screen time (9.24 h) and work/study (3.24 h).

5308 Count of user id	7500 Count of user id	26.57 Average of age	7.50 Average of daily screen time hours
3.27 Average of social media hours	2.01 Average of gaming hours	6.74 Average of sleep hours	134.26 Average of notifications per day
97.83 Average of app opens perday	9.24 Average of weekend screen time	3.24 Average of work study hours	

Figure 1: KPI cards

Visualization page. Figure 2 presents the six charts and three slicers that answer the business questions above.

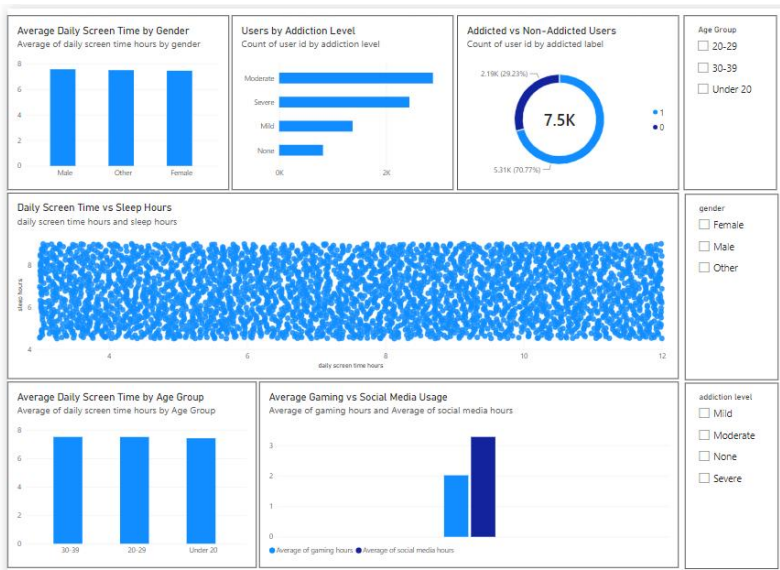


Figure 2: Data visualization page with slicers

- **Gender:** average daily screen time is almost identical (Male 7.6 h, Other 7.5 h, Female 7.4 h), so gender is not a strong differentiator.
- **Addiction levels:** Moderate is the largest group, followed by Severe, Mild and None.
- **Addicted vs not addicted:** 70.77% (5.31K) of users are addicted against 29.23% (2.19K) who are not.
- **Screen time vs sleep:** the scatter is an even cloud with a correlation of about 0.02, so higher screen time does not translate into visibly less sleep in this data.
- **Age group:** all three cohorts average about 7.5 h a day, and users spend more time on social media (3.27 h) than on gaming (2.01 h).

5. Dashboard

5.1. Dashboard Design Screenshot with Insights

The **Digital Well-Being & Addiction Insights Dashboard** shows at a glance how excessive phone use affects daily routine, sleep and stress. The design uses KPI cards at the top, horizontal bars for comparisons, pie charts for distributions and an addiction-level slicer, with a consistent colour scheme so severity groups stand out. Counsellors, faculty and health advisors can use it to spot high-risk groups and plan digital-detox sessions.

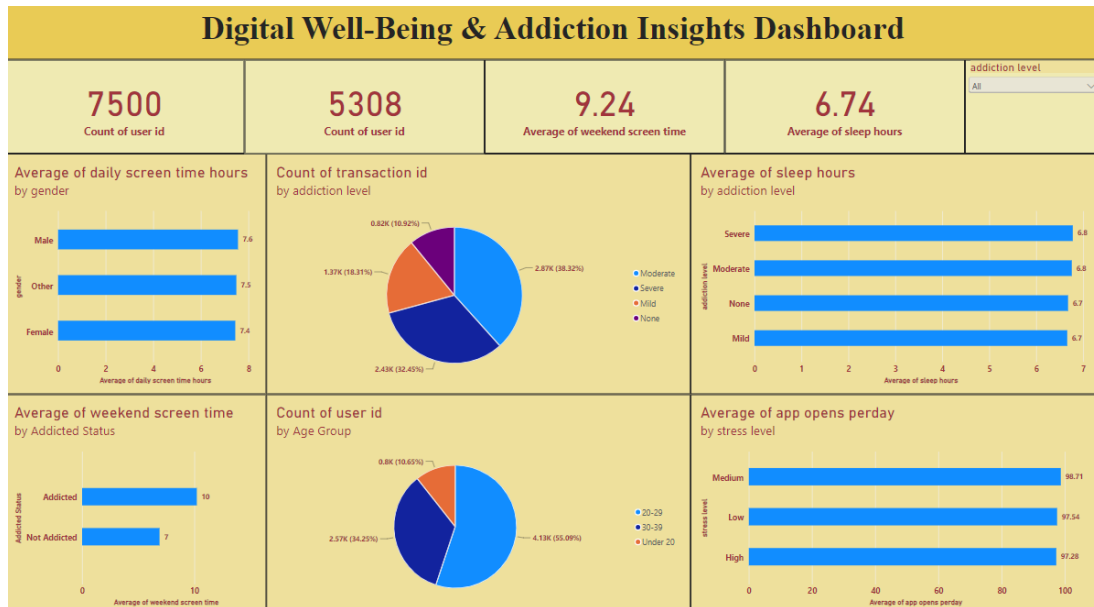


Figure 3: Digital Well-Being & Addiction Insights Dashboard

- **Users and averages:** 7,500 users assessed, 5,308 with clear dependency; average weekend screen time is 9.24 h and average sleep 6.74 h.
- **Gender:** daily screen time is 7.6 h (Male), 7.5 h (Other) and 7.4 h (Female).
- **Severity:** Moderate 38.32% (2,870), Severe 32.45% (2,430), Mild 18.31% (1,370) and None 10.92% (820).
- **Weekend gap:** addicted users spend 10 h on weekend screens vs 7 h for non-addicted users.
- **Age mix:** 20–29 is the largest group at 55.09% (4,130), then 30–39 at 34.25% (2,570) and Under 20 at 10.65% (800).
- **Sleep and stress:** average sleep is about 6.7–6.8 h at every addiction level, and app opens per day are similar across stress levels (about 97–99).

6. Report

6.1. Report Design Screenshot with Insights

Report 1: Digital Addiction & Screen Time Behaviour Analysis

Report 1 studies how daily digital habits drive dependency. Slicers for addiction level and age group, segmented bar charts and a decomposition tree let reviewers filter cohorts and trace habit patterns.

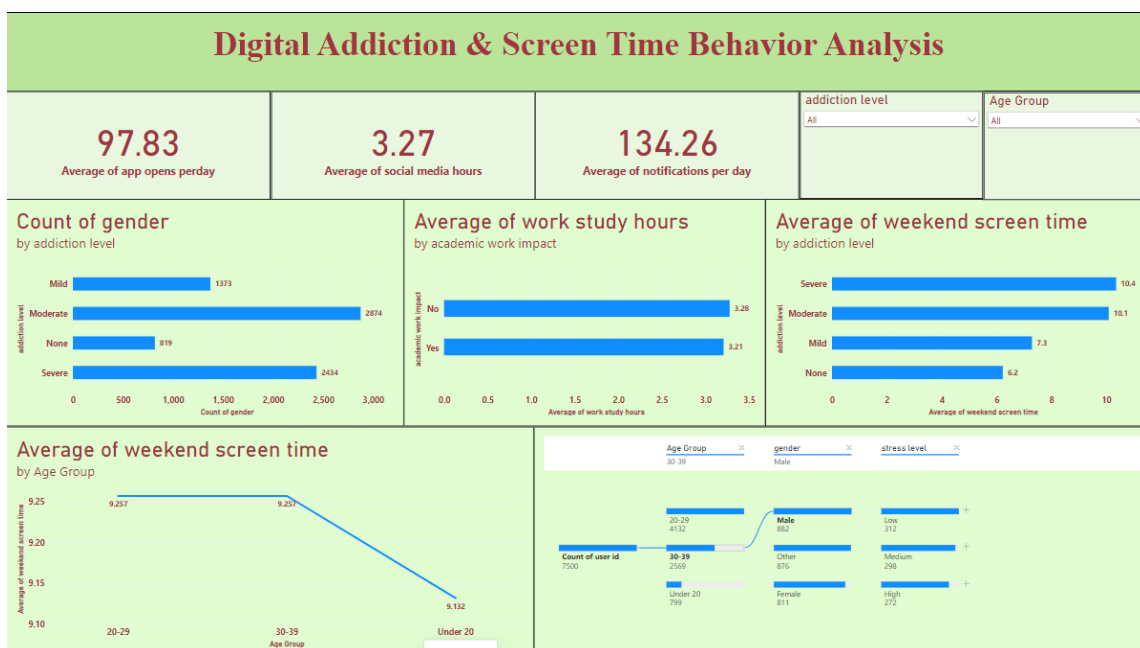


Figure 4: Report 1, Digital Addiction & Screen Time Behaviour Analysis

- **Alerts and checking:** 97.83 app opens and 134.26 notifications per day, plus 3.27 h of social media, point to continuous checking habits.
- **Addiction counts:** Moderate 2,874 and Severe 2,434 users dominate; Mild has 1,373 and None 819.
- **Weekend rise:** weekend screen time climbs with severity: 10.4 h (Severe), 10.1 h (Moderate), 7.3 h (Mild) and 6.2 h (None).
- **Age and work:** ages 20–29 and 30–39 average 9.257 h on weekends versus 9.132 h for Under 20; work/study hours are similar with impact (3.21 h) and without (3.28 h).
- **Decomposition tree:** of 2,569 users aged 30–39, 882 are male, split almost evenly across Low (312), Medium (298) and High (272) stress.

Report 2: Smartphone Addiction & Digital Well-Being Insights

Report 2 identifies the behaviours behind dependency and their effect on well-being, using donut charts, a sleep trend and the Key Influencers visual.

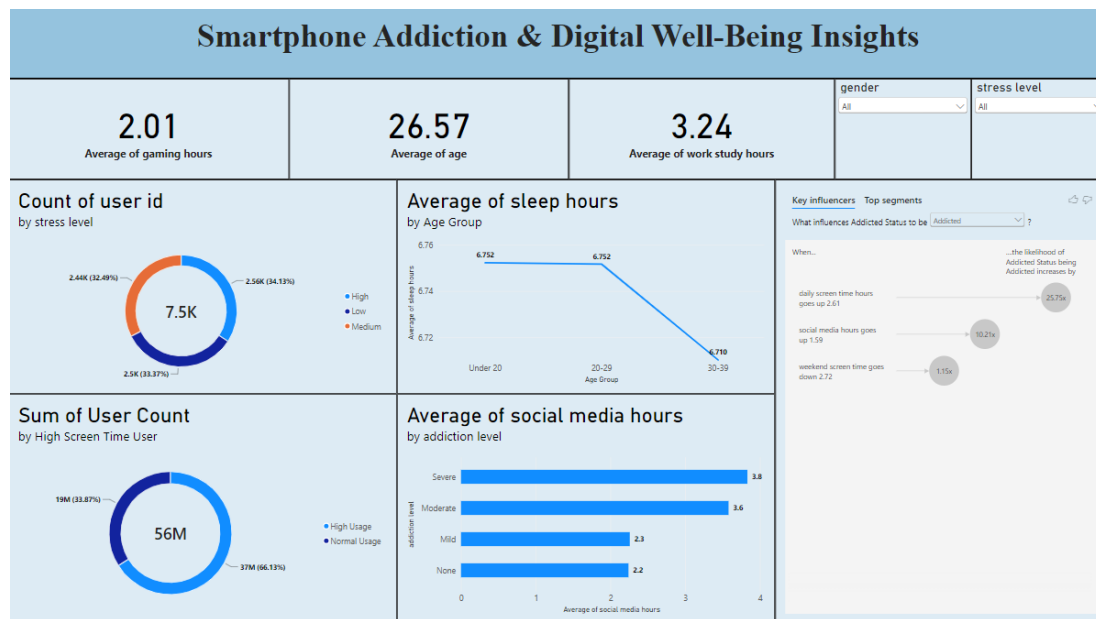


Figure 5: Report 2, Smartphone Addiction & Digital Well-Being Insights

- **Profile and stress:** average age 26.57, gaming 2.01 h and work/study 3.24 h a day; stress splits almost equally into High (34.13%), Medium (33.37%) and Low (32.49%).
- **Screen-time groups:** 66.13% of users fall in High Usage and 33.87% in Normal Usage. The donut centre reads 56M because User Count holds 7,500 on every row, so only the percentage split is meaningful.
- **Social media:** Severe users spend 3.8 h and Moderate users 3.6 h on social media, against 2.2 h for users with no addiction.
- **Sleep by age:** average sleep is 6.752 h for Under 20 and 20–29 but slips to 6.710 h for ages 30–39.
- **Key influencers:** a 2.61 h rise in daily screen time raises the likelihood of being addicted 25.75×; a 1.59 h rise in social media hours raises it 10.21×; weekend screen time adds 1.15×.

7. Performance Testing

7.1. Utilization of Data Filters

Eight slicers and two visual-level filters let stakeholders slice the data by demographic and addiction segments.

Report Page	Slicers	Slicer Fields	Other Filters
KPI page	0	—	Visual-level filter: Addicted Status = Addicted
Visualization page	3	Age Group, Gender, Addiction Level	—
Dashboard	1	Addiction Level	Visual-level filter: Addicted Status = Addicted
Report 1	2	Addiction Level, Age Group	Decomposition tree drill-down (age, gender, stress)
Report 2	2	Gender, Stress Level	Key Influencers target: Addicted Status
Total	8		2 visual-level filters

7.2. No. of Calculation Columns/Measures

Item	Count	Details
Calculated columns used in visuals	4	Age Group, Addicted Status, High Screen Time User, User Count
Additional preprocessing columns	3	Academic/work impact tag, stress category and sleep category (from the Data Quality Report)
Explicit DAX measures	0	Visuals use built-in aggregations (Count of user id, Average of hour and count columns, Sum of User Count)

7.3. No. of Visualizations

The report contains **51 visuals** across five pages (plus three text boxes used as page titles).

Visual Type	Count	Where Used
Card	21	KPI page 11, Dashboard 4, Report 1 3, Report 2 3
Clustered bar / column chart	9 / 3	Bar: Visualization 1, Dashboard 4, Report 1 3, Report 2 1; Column: Visualization page 3
Donut / pie chart	3 / 2	Donut: Visualization 1, Report 2 2; Pie: Dashboard 2
Line / scatter chart	2 / 1	Line: Report 1 and Report 2 (by age); Scatter: Visualization page
Decomposition tree / Key influencers	1 / 1	Report 1 / Report 2 (AI visuals)
Slicer	8	Visualization 3, Dashboard 1, Report 1 2, Report 2 2
Total	51	KPI page 11, Visualization 9, Dashboard 11, Report 1 10, Report 2 10

With a single 7,500-row table, no relationships and no complex DAX, the model is lightweight, so visuals and slicers refresh with negligible delay.

8. Conclusion/Observation

The analysis shows that smartphone dependency is widespread in the studied group: **70.77% of users (5,308 of 7,500) are classed as addicted**, most of them at Moderate or Severe level. The main observations are:

- **Screen time is the strongest signal.** Users average 7.50 h a day, and 66.13% are High Usage. Daily screen time and social media hours are the leading influencers of addiction.
- **Weekends amplify the habit.** Weekend screen time (9.24 h) is well above the 7.50 h daily average, reaching 10.4 h for Severe users versus 6.2 h for non-addicted users.
- **Constant interruptions.** About 134 notifications and 98 app opens a day support the problem statement that alerts drive repeated phone pickups.
- **Demographics differ little.** Gender and age groups show nearly the same daily screen time, so addiction is a behaviour pattern rather than a demographic one.
- **Sleep and stress show weak links.** Sleep stays near 6.7–6.8 h at all addiction levels and stress is split about equally in thirds, so these effects are subtle in this dataset.

Limitations: values are evenly spread across their ranges, so the results are best read as indicative patterns, not proof of cause. Absolute figures should use Count of user id rather than the User Count column.

9. Future Scope

- **Real device data:** replace the current dataset with live screen-time telemetry from Android and iOS devices.
- **Richer model:** add device, operating system and app-category dimensions and move to a star schema.
- **DAX measures:** create a Digital Addiction Risk Index and time-intelligence measures for trend tracking.
- **Predictive analytics:** train a classification model to predict addiction risk and surface it in Power BI.
- **Publishing and alerts:** publish to Power BI Service with scheduled refresh, row-level security and threshold alerts.
- **Intervention tracking:** link the dashboard to digital-detox programmes and measure before-and-after improvement.

10. Appendix

10.1. Source Code

SQL queries (file: smartphone.sql) used to explore and validate the data before reporting:

```
USE smartphone;

-- 1. View all records
SELECT * FROM smartphone_usage;

-- 2. Users and addicted users by gender
SELECT gender, COUNT(*),
       SUM(CASE WHEN addicted_label = 1 THEN 1 ELSE 0 END) AS addicted_count
FROM smartphone_usage GROUP BY gender;

-- 3. Overall averages
SELECT AVG(daily_screen_time_hours), AVG(sleep_hours), AVG(stress_level = 'High')
FROM smartphone_usage;

-- 4. Missing-value check
SELECT * FROM smartphone_usage
WHERE age IS NULL OR daily_screen_time_hours IS NULL OR addiction_level IS NULL;

-- 5. Addicted label vs stress level
SELECT addicted_label, stress_level, COUNT(*)
FROM smartphone_usage
GROUP BY addicted_label, stress_level
ORDER BY addicted_label, stress_level;

-- 6. Screen time and sleep by addiction level
SELECT addiction_level, COUNT(*), AVG(daily_screen_time_hours), AVG(sleep_hours)
FROM smartphone_usage GROUP BY addiction_level;

-- 7. Usage by academic/work impact
SELECT academic_work_impact, AVG(daily_screen_time_hours),
       AVG(social_media_hours), AVG(gaming_hours)
FROM smartphone_usage GROUP BY academic_work_impact;
```

10.2. GitHub & Project Demo Link

GitHub Repository	https://github.com/YaminiC21/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics.git
Project Demo Video	https://drive.google.com/file/d/16kxpZLgf7dKdYEl9Z1-6CsiAfBEE0N7T/view?usp=sharing
Dataset (Google Drive)	https://drive.google.com/file/d/1jaS03ZC96lXiiMtm0BwRDaus_uU1sMX1/view
Power BI File	smartphone.pbix (5 pages: KPIs, visualization, Dashboard, Report 1, Report 2)